Representation Instability in Induced Ontologies

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Abstract

Cross-embedder ontology agreement on induced clusterings is near chance (ARI ≈ 0.10). We trace the full arc from "geometry can certify ontology membership" to "geometry is unreliable for certification but useful for proposal" through 18 experiments on a 658-entry, 57-cell ontology. The dominant cause is representation instability: different embedder cones point in different directions, and no rank intervention recovers the geometry. Anisotropy is a measured covariate (mean pairwise cosine 0.53–0.56, slope -0.489 per unit), not a lever. The practical consequence is a division of labour: geometry proposes, humans dispose.

1 Introduction

Ontology induction — discovering taxonomic structure from data — is a founding promise of representation learning. If embeddings capture semantic similarity, then clustering embeddings should recover ontological structure. We test this on a 658-entry, 57-cell industrial ontology (5 domains $\times$ 14 modes), using three vector sets over the same entries: BGE-base with name+hints text, BGE-base with name-only text, and Nomic-embed.

The result is a settled negative: cross-embedder adjusted Rand index (ARI) is 0.108–0.178, near chance. The negative is now well-understood: the disagreement has a measurable covariate (anisotropy) and a known mechanism (cone orientation differs by embedder). But the negative does not imply uselessness: geometry remains a powerful proposal mechanism.

2 Experiments

2.1 Cross-embedder ARI

Cross-embedder clustering ARI was measured with spherical k-means at $k = 57$ (the number of true cells), with ARI computed against the true cell assignments:

The covariate's predicted direction holds: the least anisotropic set (Nomic, 0.527) has the best fit to true cells (0.080), and the slope of fit ARI against anisotropy is -0.489 per unit. Three points cannot separate a slope from noise; what is reproducible is the ARI itself.

Vector set	Anisotropy	Fit ARI	Cross-embedder ARI
BGE name+hints	0.558	0.062	0.108
BGE name-only	0.562	0.066	0.165
Nomic-embed	0.527	0.080	0.178

Table 1: Anisotropy (mean pairwise cosine), fit ARI against true cells, and cross-embedder clustering ARI. All near chance (ARI = 0).

2.2 Injection realism

The central confound was identified in E14: the injection policy determines the verdict.

Injection policy	AUC
Random cell (original)	0.880
Same domain, different mode	0.827
Same mode, different domain	0.771
Nearest alternative cell	0.506
Real misfilings (E16)	0.519

Table 2: Injection realism. Making the injected error plausible drops the scorer to chance, and closes the 0.410/0.866 gap this track carried unexplained throughout.

The scorer detects *distance moved*, not wrongness. A random injection displaces an entry 0.125 below its home similarity; a plausible error, 0.029.

2.3 Structural scorers

Five scorers were compared under four injection policies (E15):

Scorer	Random inj.	Nearest inj.
COSINE	0.882	0.506
NAME-ONLY	0.866	0.597
CONSTRAINT	0.675	0.612
ANCHORVOC	0.654	0.584
ANCESTRY	0.627	0.566

Table 3: Structural scorers under two injection policies. Under random injection, cosine leads by 0.207. Under realistic injection, cosine trails CONSTRAINT by 0.106. Stage 12's comparison was invalid.

Under random injection, cosine leads. Under realistic injection, it trails. This does not show the structural scorers work — the best is 0.612, not a usable certifier — but it shows the comparison that established "plain cosine beats every structural alternative" was an artifact of the injection policy.

2.4 ANCHORVOC and the length artifact

ANCHORVOC (lexical overlap with the cell's authored anchor hints) scored 0.664 on real misfilings (E16), beating cosine's 0.519 by +0.145. It was the only scorer that won on real data. It was also an entry-length artifact: entry length alone reached 0.619 of the 0.664, and no vocabulary effect survived controlling for it (E18).

2.5 Propose, do not certify

The one positive that survives its construction-matched null is E21: geometry as a proposer, not a judge. Top-3 containment of the rater's own cell achieved 0.712 against a permuted-label null of 0.136, with the learning curve still climbing at 436 decisions. The geometry did not get better — a number filed as a failure (0.447 top-1) is the same number that makes a usable proposer once the output is three candidates instead of one.

3 The arc

The research lineage is:

1. **Claim:** geometry can certify ontology membership.
2. **E1:** cross-embedder divergence is at chance.
3. **E2–E5:** the taxonomy is not the constraint; the grid is not the constraint.
4. **E14:** the injection policy determines the verdict.
5. **E15:** the structural scorers do not survive the correction.
6. **E16:** all geometric scorers are at chance on real misfilings.
7. **E17–E18:** ANCHORVOC is an entry-length artifact.
8. **Verdict:** no tested mechanism detects a real misfiling. The central negative stands, now for the right reason.
9. **E21:** geometry can propose. The task formulation changes, not the representation.
10. **E88–E90:** anisotropy is real but not a lever; rank intervention fails.

Four experiments each correcting the last, ending at the position the track held before E14 — for entirely different and much better reasons.

4 Discussion

The instrument, not the mechanism. The cross-embedder refutation is reproducible (a new clustering implementation lands in the same band). But the instrument — the corpus, the embedding model, the clustering algorithm — dominates the mechanism. Geometry is unreliable for certification because the representation is unstable, not because the ontology is absent.

The division of labour. The propose/certify distinction is the finding. Geometry converts human decisions into leverage over the remaining ones (top-3 0.712 vs. null 0.136). It does not replace the human. The practical rule: sell the filter, not the judgement.

Limits. One corpus (658 entries, 57 cells), one embedder pair (BGE-base vs. Nomic-embed), one clustering algorithm (spherical k-means). The externalization to WordNet/SUMO/DOLCE with 3+ embedders is the highest-priority replication. The anisotropy slope rests on three points and we treat it as exploratory, not as an established relationship.

Reproducibility and provenance

Every numerical claim in this paper now resolves to a stored result file.

The headline results — the anisotropy measurements, the fit ARIs, and the cross-embedder ARIs — come from `research/mapper/e88_results.json` and `research/mapper/e90_results.json`, produced by `e88_anisotropy_covariate.py` and `e90_rank_intervention.py`.

The comparison numbers from the E14–E18 chain were previously unreproducible: those four experiments shipped only as printing scripts and their output was never saved. They have since been re-run from the surviving corpus and ground truth, and the recovered values are stored in `research/e15_e18_rerun.json` (raw output in `research/e15_e18_rerun_raw.txt`, parsed by `research/papers/parse_e1518.py`). The re-run reproduced the originally reported figures:

- the random-injection AUCs (COSINE 0.882, NAME-ONLY 0.866, CONSTRAINT 0.675, ANCHORVOC 0.654, ANCESTRY 0.627),
- the nearest-cell injection fall (COSINE 0.506, CONSTRAINT 0.612),
- the real-misfiling battery (COSINE 0.519, NAME-ONLY 0.508, CONSTRAINT 0.536, ANCHORVOC 0.664, ANCESTRY 0.534, base rate 80.1%),
- the length control that closes ANCHORVOC (entry length alone 0.619; normalised ANCHORVOC vs. length-only +0.048 [−0.019, +0.116], not significant).

One number deserves emphasis because the re-run changed its status from assertion to measurement: E18's port check shows the Python scorer reproduces the original Rust values for CONSTRAINT ($r = 0.943$) and ANCESTRY ($r = 0.976$) but *not* for COSINE ($r = 0.197$). The divergence is expected — the original used MiniLM, this corpus uses BGE — and it means the structural scorers are embedder-independent while geometry is not. That is a stronger version of this paper's thesis and it is now measured rather than inferred.